

Ahsan Habib Akash

Ph.D. Student in Electrical Engineering · Mechanistic Interpretability & Safety of Vision–Language Models

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Research Interests

Mechanistic interpretability, evaluation, and post-training of vision–language models (VLMs): whether reinforcement-learning post-training yields genuine visual grounding or amplifies language priors; diagnostic benchmarks that expose failures hidden by aggregate metrics; and trustworthy, fair VLMs for medical imaging.

Education

West Virginia University

Ph.D. in Electrical Engineering | Specialization: Computer Vision

Advisor: Dr. Prashnna K. Gyawali

Morgantown, WV

Jan 2024 – Dec 2028 (expected)

Bangladesh University of Engineering and Technology (BUET)

B.Sc. in Electrical and Electronic Engineering

Dhaka, Bangladesh

Feb 2017 – May 2022

Publications

CONFERENCE & WORKSHOP PAPERS

When Medical VLMs Stop Understanding: MedTEC-Bench for Probing Semantic Specificity Dec 2026

Ahsan Habib Akash, Alina Devkota, Donald Adjeroh, Binod Bhattacharai, Prashnna K. Gyawali

Advances in Neural Information Processing Systems (NeurIPS 2026), Evaluations & Datasets Track · Poster

Two is Better than One: A Collapse-free Multi-Reward RLIF Training Framework Dec 2026

Shourov Joarder, Diganta Sikdar, Ahsan Habib Akash, Binod Bhattacharai, Prashnna K. Gyawali

Advances in Neural Information Processing Systems (NeurIPS 2026), Main Track · Poster · arXiv:2605.22620

Addressing Bias in VLMs for Glaucoma Detection Without Protected Attribute Supervision Oct 2025

Ahsan Habib Akash, Greg Murray, Annahita Amireskandari, Joel Palko, Carol Laxson, Binod Bhattacharai, Prashnna K. Gyawali

3rd Workshop on Data Engineering in Medical Imaging (DEMI), MICCAI 2025 · arXiv:2508.09087

Study of Multi-class Arrhythmia Detection and Classification from ECG Data using Deep Learning Dec 2023

Ahsan Habib Akash, R. Ahamed

26th International Conference on Computer and Information Technology (ICCIT), Cox's Bazar, Bangladesh ·

doi:10.1109/ICCIT60459.2023.10441043

SignBD-Word: Video-Based Bangla Word-Level Sign Language and Pose Translation Jul 2023

A. Sams, Ahsan Habib Akash, S. M. M. Rahman

14th International Conference on Computing, Communication and Networking Technologies (ICCCNT), Delhi, India ·

doi:10.1109/ICCCNT56998.2023.10306914

Activity Classification from First-Person Office Videos with Visual Privacy Protection Oct 2022

Partho Ghosh, Md Abrar Istiak, Nayeef Rashid, Ahsan Habib Akash, Ridwan Abrar, Ankan Ghosh Dastider, Asif Shahriyar Sushmit, Taufiq Hasan

The Fourth Industrial Revolution and Beyond, Springer Nature, Singapore, pp. 157–169

JOURNAL ARTICLES

Robust CNN Multi-Nested-LSTM Framework with Compound Loss for Patch-Based Multi-Push Ultrasound Shear Wave Imaging and Segmentation 2024

Md. Jahin Alam, Ahsan Habib Akash, Muyinatu A. Lediju Bell, Md. Kamrul Hasan

Physics in Medicine and Biology

PREPRINTS

SW-ViT: A Spatio-Temporal Vision Transformer Network with Post Denoiser for Sequential Multi-Push Ultrasound Shear Wave Elastography May 2025

Ahsan Habib Akash, Md. Jahin Alam, Md. Kamrul Hasan

arXiv:2505.18865

Research Experience

West Virginia University

Morgantown, WV
Jan 2025 – Present

Graduate Research Assistant | Advisor: Dr. Prashnna K. Gyawali

- **Mechanistic interpretability of VLMs:** testing whether RL post-training induces genuine visual grounding or amplifies language priors, using activation patching, causal tracing, and counterfactual evidence perturbation (TransformerLens); locating attention heads responsible for multimodal grounding and analyzing how visual tokens enter the decoder’s residual stream.
- **Evaluation & benchmarking:** built MedTEC-Bench, a diagnostic benchmark with a five-level probing taxonomy that exposes semantic-specificity failures in medical VLMs hidden by aggregate accuracy (first author, NeurIPS 2026 E&D).
- **RL post-training:** co-developed a collapse-free, multi-reward reinforcement-learning-from-internal-feedback (RLIF) framework with GRPO-style fine-tuning that improves visual faithfulness without external reward supervision (NeurIPS 2026).
- **Interpretable & fair medical AI:** language-anchored concept discovery and sparse autoencoders for human-interpretable concept directions; unsupervised bias mitigation for VLM-based glaucoma screening without protected-attribute labels (DEMI @ MICCAI 2025).

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh
Dec 2022 – Dec 2023

Research Assistant, DSP Research Lab | Advisor: Dr. Md. Kamrul Hasan

- Collaborated with Fujifilm (Japan and USA) on deep learning for ultrasound shear-wave elastography acquisition.
- Built multi-resolution spatio-temporal vision-transformer models (SW-ViT) with a synthetic-to-real pipeline: COMSOL Multiphysics simulation for pre-training, CIRS phantom data for fine-tuning.
- Led medical image segmentation and classification projects, resulting in a first-author preprint and a co-authored journal article (*Physics in Medicine and Biology*, 2024).

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh
Sept 2022 – Dec 2022

Research Assistant, Telecommunication Lab | Advisor: Dr. Md. Farhad Hossain

- Developed deep-learning models for anomaly detection in network traffic.

Bangladesh University of Engineering and Technology (BUET)

Dhaka, Bangladesh
Aug 2021 – 2022

Undergraduate Research | Advisor: Dr. S. M. Mahbubur Rahman

- Video-based Bangla sign-language translation (SignBD-Word) with I3D, SlowFast, and CNN–LSTM recognition models (ICCCNT 2023).

Selected Course Projects (West Virginia University, 2024 – Present)

Multi-stage GAN for satellite image super-resolution · residual diffusion denoising for universal image restoration · attention graph neural network for word-level ASL recognition · windowed graph attention for image compression and super-resolution.

Awards & Achievements

2nd place, MICCAI 2025 Multi-Camera Robust Diagnosis of Fundus Diseases (MuCaRD) Challenge	2025
Top 8, AI for Bangla 2.0	2023
Top 6, Robi Datathon 2.0	2022
1st place, IEEE VIP Cup @ ICIP 2020 — real-time vehicle detection and tracking with fisheye cameras	2020
5th place, IEEE Signal Processing Cup @ ICASSP 2020 — real-time anomaly detection in ground/aerial systems	2020
3rd place, IEEE VIP Cup @ ICIP 2019 — activity recognition from body cameras	2019

Technical Skills

ML research	Mechanistic interpretability (TransformerLens, activation patching, sparse autoencoders), RL post-training (GRPO), benchmark design and evaluation, bias mitigation, federated learning
Languages & tools	Python, PyTorch, TensorFlow, OpenCV, C/C++, MATLAB, COMSOL Multiphysics
Compute	HPC clusters (PSC Bridges-2, NCSA DeltaAI, CSCS Clariden), MLOps